

Chapter 44: Management of Obstetrical Hemorrhage

— Revision Notes

Williams Obstetrics 26e, pp. 2000–2035 of the PDF. High-yield numbers are in **bold**. The two tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 6 figures are included from the PDF.

Big picture: Visual estimates of blood loss are notoriously inaccurate: true loss is often **2–3× the estimate**, and some or all of it may be concealed. Management = **find and stop the source**, while you **resuscitate with crystalloid and blood, replace procoagulants** (dilutional and consumptive coagulopathy are treated the same way), and escalate to **surgical or radiological** control. Hourly **urine output** is one of the most important vital signs.

Recognizing the Severity of Hemorrhage

- Peripartum loss includes the **pregnancy-augmented blood volume**.
- **Rule of thumb:** once pregnancy hypervolemia is lost, each **3-volume-percent fall in hematocrit $\approx$ 500 mL** blood loss.
- ⚠ With ongoing bleeding, the hematocrit measured in the delivery, operating or recovery room is at its **maximum**; the nadir depends on how fast crystalloid and blood are given.
- If loss seems more than average → check hematocrit and observe closely for deterioration. (ACOG blood-loss assessment: Ch 42.)
- **Urine output** reflects renal and other vital-organ perfusion: keep **≥ 30 mL/h, preferably ≥ 50 mL/h**. Diuretics are seldom indicated during active bleeding.
- Always ask whether there are **enough procoagulants** for clot formation: many severe hemorrhages are complicated by **DIC**.

1. Management of Hemorrhage

Hypovolemic shock: pathophysiology

- **Early:** ↓ MAP, stroke volume, cardiac output, CVP and PCWP; ↑ arteriovenous O₂ difference (more extraction) but total O₂ consumption falls.
- **~70% of blood volume is in the venules** (capacitance vessels, humorally controlled) → catecholamines ↑ venular tone = **autotransfusion**. Also ↑ HR, SVR, PVR and contractility.
- **Autoregulation** redistributes flow **to heart, brain and adrenals**, away from kidneys, splanchnic bed, muscle, skin and **uterus**.
- ⚠ **Volume deficit $> \sim 25\%$ → compensation fails**; small further losses cause **rapid deterioration**.
- Then: tissue hypoxia and **metabolic acidosis** → vicious cycle of vasoconstriction, ischemia and cell death. Leukocyte activation → **endotheliopathy** and platelet aggregation → microvascular occlusion. Preeclampsia or sepsis add capillary leak and can **incite DIC**.
- Fluid and electrolyte shifts (Na⁺ and water into skeletal muscle, K⁺ loss) → **both extracellular fluid and intravascular volume must be replaced**. Survival is better with **blood + crystalloid** than with blood alone.

Fluid resuscitation

- **Simultaneously** identify the source and resuscitate (algorithm: Fig 42-3).
 - Undelivered: restoring volume helps mother and fetus and prepares for emergent delivery.
 - Postpartum: immediately look for **atony, retained placental fragments, genital tract lacerations**.

- **One or two large-bore IV lines**, rapid crystalloid, order blood; ready the OR; assemble surgical and anesthesia teams (plus neonatal team if delivery is imminent).
- Only **20% of crystalloid remains intravascular after 1 h** → give **2–3× the estimated blood loss** initially.

Crystalloid vs colloid

- **Little or no difference in mortality** (Cochrane; SAFE trial, ~7000 patients).
- Parkland: **crystalloid + blood**. Saline or **balanced solutions** (Ringer lactate, Plasma-Lyte, whose electrolytes resemble plasma). Balanced were superior in some, not all, studies. Excess saline → theoretical **hyperchloremic acidosis**.

2. Blood Replacement

Type and screen vs crossmatch

- Parkland does a **type and screen on every woman admitted to labor and delivery** (PPH prediction is poor).
- **Screen** = maternal serum + standard reagent red cells (common significant antigens). **Crossmatch** = uses actual donor red cells.
- Only **0.03–0.07%** of antibody-negative screens later have antibodies → screened blood **rarely** causes adverse effects.

Transfusion thresholds

- Cardiac output does not fall substantially until **Hb ~7 g/dL (Hct ~20%)**. Military target **Hct 21%**, where most experts consider transfusion.
- **Williams: with ongoing obstetrical hemorrhage, transfuse rapidly when Hct <25%**. Modify for undelivered fetus, imminent surgery or expected ongoing loss, hypoxia, vascular collapse.
- Nonpregnant evidence: restrictive (**Hb >7**) vs liberal (10–12 g/dL) → 30-day mortality **19% vs 23%** (similar). Septic shock: 7 vs 9 g/dL targets gave similar mortality.
- Units needed depend on **body mass** and expected further loss.

Components and ratios

- Whole blood is rarely available → most get **PRBCs + crystalloid + components**, often through a **massive transfusion protocol (MTP)**.
- Trauma data (massive transfusion = **≥10 units**): best survival as **plasma:RBC approaches 1:1.4**; worst at **1:8**.
- After **~5 units RBC** → check **platelets, clotting studies and fibrinogen**.

Target in obstetrical hemorrhage	Action
Platelets >50,000/μL	Platelet concentrates
Fibrinogen <150 mg/dL , or prolonged PT/PTT with surgical bleeding	FFP 10–15 mL/kg or cryoprecipitate

- **Parkland escalation (no whole blood): 2 units PRBC** → if bleeding continues, **2 PRBC + 2 FFP** automatically → then **activate MTP**.

Table 44-1. Blood Products Commonly Transfused in Obstetrical Hemorrhage

Product	Volume per Unit	Contents per Unit	Effect on Hemorrhage
Whole blood	About 500 mL; Hct ~40 percent	RBCs, plasma, 600–700 mg fibrinogen, no platelets	Restores blood volume and fibrinogen, increases Hct 3–4 volume percent per unit
Packed RBCs	250–300 mL; Hct ~55–80 percent	RBCs, minimal fibrinogen, no platelets	Increases Hct 3–4 volume percent per unit
FFP	About 250 mL; 30-minute thaw	Colloid, 600–700 mg fibrinogen , no platelets	Restores circulating volume and fibrinogen
Cryoprecipitate	About 15 mL, frozen	One unit has 200 mg fibrinogen , other clotting factors, no platelets	15–20 units or 3–4 g will increase baseline fibrinogen ~150 mg/dL
Platelets	About 50 mL, stored at room temperature	One unit raises platelet count ~5000/μL ; single-donor apheresis bag preferable	6–10 units transfused; single-donor bag preferable to raise platelets ~50,000/μL

FFP = fresh-frozen plasma; Hct = hematocrit; RBCs = red blood cells.

The whole-blood volume prints as "About 50 mL", an obvious typo for ~500 mL (a standard unit). The platelet unit prints as " μ /L" and should be / μ L. The text says one single-donor apheresis bag raises platelets by ~20,000/ μ L, while the table's last cell gives ~50,000/ μ L for the full 6–10-unit transfusion.

Whole blood and packed red cells

- **Whole blood is ideal** for severe obstetrical hemorrhage: shelf life **40 days**; **70%** of cells function ≥ 24 h; **one unit** $\uparrow$ **Hct 3–4%**. It replaces **fibrinogen** and other factors, treats hypovolemia, and means **fewer donor exposures**.
- Parkland (>66,000 deliveries): whole blood $\rightarrow$ **less severe morbidity** (renal failure, ARDS, pulmonary edema, hypofibrinogenemia, ICU admission, death) than PRBC + components.
- **PRBC**: Hct **55–80%**; one unit $\uparrow$ Hct **3–4%** (depends on size).

Dilutional coagulopathy

- Massive crystalloid + PRBC deplete platelets and factors $\rightarrow$ **clinically indistinguishable from DIC**.
- **Thrombocytopenia is the most frequent coagulation defect** after blood loss and multiple transfusions.
- PRBCs contain only tiny amounts of soluble factors; **stored whole blood lacks platelets and factors V, VIII and XI** $\rightarrow$ hypofibrinogenemia, $\uparrow$ PT/PTT.
- Distinguishing dilutional from consumptive coagulopathy is confusing, but **treatment is the same**.

Platelets

- Transfuse with ongoing hemorrhage if **<50,000/ μ L**. Without surgery, bleeding is rare at **$\geq 10,000/\mu$ L**.
- **Single-donor apheresis bag** = equivalent of **6 donor units** $\rightarrow$ raises count **~20,000/ μ L**. Otherwise **6–8** individual units.
- Donor plasma must be **compatible with recipient red cells**; **D-negative women get platelets only from D-negative donors** (some red cells are always transfused).

Fresh-frozen plasma

- Takes **~30 min to thaw**. Contains **all stable and labile factors, including fibrinogen** $\rightarrow$ for consumptive or dilutional coagulopathy.
- **⚠ Not a volume expander** without a specific factor deficiency. Consider with fibrinogen **<150 mg/dL** or abnormal PT/PTT.
- **Liquid plasma** (never frozen, stored at **1–6°C** up to **40 days**) compares favorably; not widely available.

The extracted text prints "1 to 60 C", which is 1–6°C.

Cryoprecipitate and fibrinogen concentrate

- One unit of cryo (from one unit of FFP, **10–15 mL**) has **≥200 mg fibrinogen + factor VIII:C, VIII:vWF, factor XIII, fibronectin**. Usually given as a pooled "bag".
- **Ideal fibrinogen source** when levels are dangerously low and incisions ooze.
- **Most commonly wasted product** in labor and delivery (~**17%** of ordered cryo).
- **Virus-inactivated fibrinogen concentrate** (RiaSTAP, Fibryna): each **1 g ↑ fibrinogen ~40 mg/dL**; used for congenital hypofibrinogenemia.

The text says a cryo pool comes from "8 to 120 donors". The upper figure is probably a typo for about 12.

Recombinant activated factor VII (rFVIIa, NovoSeven)

- Binds exposed **tissue factor** at the injury → thrombin → platelet and cascade activation. Used in surgery, trauma and obstetrics; in many trauma MTPs.
- ⚠ **Ineffective if fibrinogen <50 mg/dL or platelets <30,000/μL.**
- Concerns: **arterial (and venous) thrombosis**; only **marginal efficacy**.

The text reports arterial thromboembolism in "55 percent" across 35 RCTs (Levi, 2010). That study actually found about 5.5%, so "55" is probably a misprint.

Tranexamic acid (TXA)

- **Antifibrinolytic**: binds plasminogen reversibly → stops plasmin from binding fibrin → clot persists.

Trial / source	Finding
WOMAN (PPH after vaginal birth or at cesarean; developing countries)	1 g IV TXA → hemorrhage deaths 1.2% vs 1.7% → WHO recommends
Prophylaxis after vaginal delivery (Sentilhes 2018)	No ↓ PPH
TRAAP2 (prophylaxis at cesarean)	↓ red-cell transfusion
Meta-analysis / Cochrane	Must be given early after bleeding onset to ↓ mortality
ACOG	Not recommended for prophylaxis or first-line treatment; if used, give within 3 h of delivery

Massive transfusion protocols

- Start once **4–5 units PRBC are given within ~2 h**. Speeds product delivery and **avoids dilutional coagulopathy**. Some include rFVIIa, TXA or prothrombin complex concentrate.
- Evidence of benefit in obstetrics is limited (mostly trauma data, some observational obstetric studies).

Table 44-2. Parkland Hospital Obstetrical Massive Transfusion Protocol

Round No.	PRBC 5 Units	FFP 3 Units	Plts 6-Pack	Cryo 1 Unit
1	X	X	—	—
2	X	X	X	—
3	X	X	—	X
4	X	X	X	—
5	X	X	—	—
6	X	X	X	X
7	X	X	—	—
8	X	X	X	—

Cryo = cryoprecipitate; FFP = fresh-frozen plasma; Plts = platelets; PRBC = packed red blood cells.

Memory hook: every round = **5 RBC + 3 FFP**; platelets on **even** rounds; cryo on rounds **3 and 6**.

Viscoelastic assays (TEG, ROTEM)

- Point-of-care whole-blood tests of **clot formation, strength and fibrinolysis**; used in trauma, liver transplant and cardiac surgery. Pregnancy reference ranges confirm **hypercoagulability**.
- Limits: less useful in **torrential** bleeding (need minute-by-minute data), **cannot detect primary hemostasis disorders**, and are easily **misinterpreted** by untrained staff.
- Williams: needs more study; more useful **after initial management / in ICU recovery** than during acute hemorrhage.

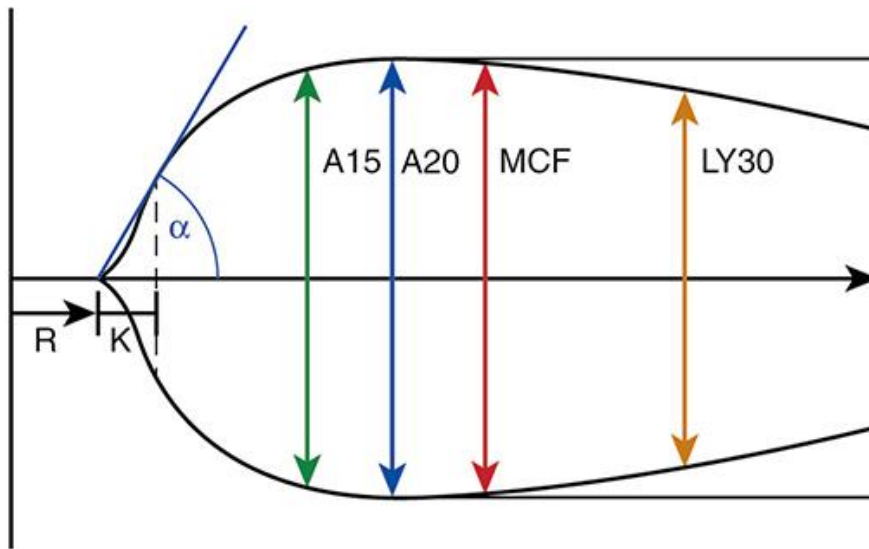


Figure 44-1. Normal TEG: R = time to clot start; K = to 20-mm amplitude; α = speed of fibrin build-up; A15/A20 = amplitude at 15/20 min; MCF = maximum clot firmness; LY30 = % lysis at 30 min.

Cell salvage and autologous transfusion

- **Preoperative autologous donation: disappointing** (exceptions: rare blood type or unusual antibodies).
- **Intraoperative cell salvage is safe** in obstetrics and helps women who **decline transfusion**. Earlier fear of amniotic fluid embolism was **not borne out** (no AFE in recent studies).
- RCT (3028 cesareans at risk): donor transfusion **2.5% vs 3.5% (not significant)**. Routine set-up in another study → lower donor transfusion.

Transfusion complications

Complication	Key facts
Acute hemolytic reaction (ABO incompatibility)	Usually clerical error (mislabeling, wrong patient): ~1 in 14,000 units . Fever, hypotension, tachycardia, dyspnea, chest/back pain, flushing, anxiety, hemoglobinuria → DIC, AKI, death. Stop transfusion , treat hypotension and hyperkalemia , induce diuresis, alkalinize urine
TACO	Most common of the two; incidence ~1%
TRALI	≥1 in 12,000 ; HLA and neutrophil antibodies in donor plasma
TACO + TRALI	Most common causes of transfusion-related death . Severe dyspnea, hypoxia, pulmonary edema within 6 h ; supportive care ± ventilation
Bacterial (red cells)	Unusual (refrigeration): <i>Yersinia</i> , <i>Pseudomonas</i> , <i>Serratia</i> , <i>Acinetobacter</i> , <i>E coli</i>
Bacterial (platelets)	Stored at room temperature : 1 in 1000–2000 units contaminated; sepsis death 1 in 17,000 single-donor platelets and 1 in 61,000 apheresis packs
HIV, hepatitis C	1 in 1–2 million units
Hepatitis B	<1 in 100,000
CMV	Infected leukocytes often transfused (high prevalence)
Others	West Nile, parvovirus B19, HTLV-1, toxoplasmosis (slight); Zika (all donations now tested); SARS-CoV-2 hypothetical

3. Obstetrical Coagulopathies

Terms

- **Consumptive coagulopathy** = actual consumption of procoagulants inside the vessels.
- **Dilutional coagulopathy** = massive **loss** of procoagulants through hemorrhage.
- Consumptive coagulopathy → systemic intravascular activation that overwhelms natural anticoagulants → widespread fibrin deposition → **multiorgan failure**.

Pregnancy-induced coagulation changes (procoagulant state)

- ↑ **Fibrinogen (I), VII, VIII, IX, X**.
- ↑ **Plasminogen**, but also ↑ **PAI-1 and PAI-2** → **plasmin activity falls** until after delivery.
- **Platelets** ↓ **~10%**, with ↑ activation.
- ↑ fibrinopeptide A, β-thromboglobulin, platelet factor 4, **FDPs/D-dimers**; ↓ **protein S**.
- Net: hypercoagulability, ↓ fibrinolysis and **compensated intravascular coagulation** that may maintain the uteroplacental interface.

The extracted text lists factors "VII, VII, IX, and X"; the duplicate is presumably VIII.

DIC in pregnancy

- Incidence **0.03–0.35%**. **Significant coagulopathy in virtually all abruptions and amniotic fluid embolism**.
- Less recognized activation: sepsis, thrombotic microangiopathies, AKI, **acute fatty liver**, severe preeclampsia, HELLP.
- 49 severe cases: transfusion **59%**, hysterectomy **18%**, dialysis **6%**, **3 maternal deaths**, PMR **30%**.
- DIC was the **2nd most common severe maternal morbidity indicator** (2010–2011) and was associated with **~¼ of maternal deaths**, yet as a **sole cause** it accounts for only **0.2%** of US pregnancy-related deaths.

Normal coagulation: cell-based (tissue factor) model

- **Tissue factor** (membrane glycoprotein) is the **principal initiator**, not the old "waterfall".

- **TF-VIIa** complexes → activate **IX and X** → **Xa** starts clotting. The "intrinsic" pathway **amplifies**: initial thrombin directly activates **factor XI** (feedback loop) → fibrin.
- TF is abundant in the **brain, lungs, placenta and amniotic fluid**.
- Balanced by **fibrinolysis** (plasminogen → plasmin) → **fibrin split products incl. D-dimers**.

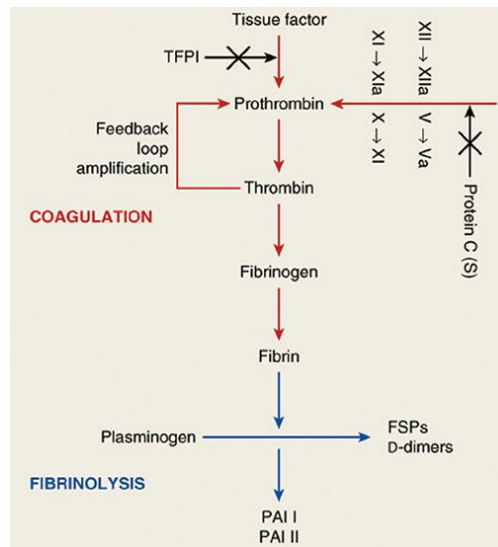


Figure 44-2. Coagulation (tissue factor → thrombin with feedback amplification → fibrin) balanced by fibrinolysis (plasminogen → FSPs/D-dimers). Inhibitors: TFPI, protein C/S, PAI-1/2. (The figure's "X → XI" label presumably means X → Xa.)

Pathological coagulation

- Disease releases **tissue factor** from subendothelium and stimulated monocytes → endothelial cytokines → **diffuse coagulation**. Clinically important once factors and platelets are depleted.
- **Purpura fulminans**: severe, often lethal. **Microthrombi** → **skin necrosis** (± vasculitis); burn-unit debridement; rarely **four-limb amputation**.
 - Usually **sepsis in women with heterozygous protein C deficiency**. **Homozygous protein C or S deficiency** → **fatal neonatal purpura fulminans**.

Inciting conditions

Condition	Coagulopathy
Placental abruption	Most common cause of severe consumptive coagulopathy in obstetrics
Preeclampsia / eclampsia / HELLP	Endothelial activation; DIC in ~10% of HELLP ; severity correlates with thrombocytopenia and FDPs, but seldom clinically worrisome
Acute fatty liver	Intravascular coagulation + ↓ procoagulant synthesis
Amniotic fluid embolism	IV embolization of (meconium-laden) amniotic fluid → cardiorespiratory collapse + profound DIC (SMFM checklist; Ch 42)
Sepsis	Seldom massive alone; <i>E coli</i> (pyelonephritis, puerperal infection) usually mild. Exceptions: exotoxins of group A Streptococcus, S aureus, Clostridium perfringens, C sordellii, C novyi
Septic abortion	Can incite coagulation and worsen bleeding
Late 2nd-trimester D&E	1.6% (1249 cases; ⅓ for fetal demise)
Hypertonic-solution abortion	Thromboplastin release from placenta, fetus, decidua (rarely used now)
Retained dead fetus	Most labor within 2 wk ; gross disruption rarely before 4 wk ; after 1 month ~¼ develop consumptive coagulopathy

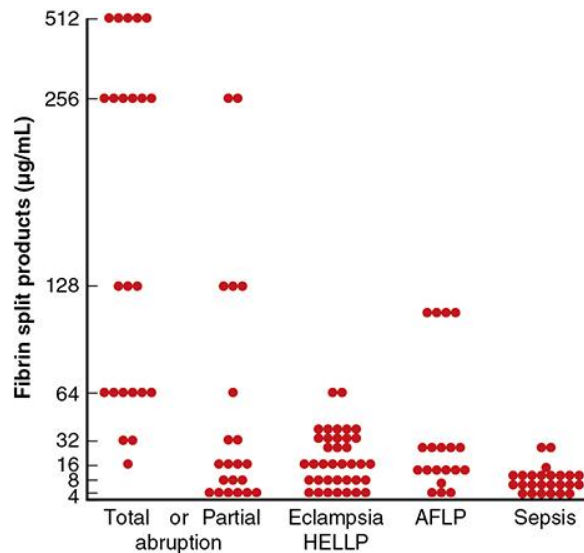


Figure 44-3. Fibrin split products are highest with total abruption (up to 512 µg/mL), lower with partial abruption, eclampsia/HELLP and AFLP, and lowest with sepsis.

Diagnosis

- **"Bioassay"** is excellent: bleeding at sites of minor trauma (venipuncture, shaving nicks, catheterization); spontaneous gum, nose or GI bleeding. **Petechiae or purpura** at cuff or tourniquet sites → thrombocytopenia. **Surgery is the ultimate bioassay** (generalized oozing).

Test	Interpretation
Fibrinogen	Late pregnancy 300–600 mg/dL , so even major consumption (e.g., 600 → 250) may still clot. Clinically important threshold ~150 mg/dL . Serious hypofibrinogenemia <50 mg/dL
Clot observation test	Whole blood in a glass tube: with low fibrinogen the clot is soft and doesn't retract ; platelet-driven retraction follows over ~30 min. No retraction within ~1 h → thrombocytopenia likely
D-dimer / FDPs	Almost always ↑ in significant consumptive coagulopathy; level does not correlate with outcome in obstetrics
Platelet count	Confirms thrombocytopenia; qualitative dysfunction possible with severe preeclampsia
PT / PTT	Prolonged by very low fibrinogen, depleted procoagulants, or large amounts of FDPs
TEG / ROTEM	Adjuncts to guide replacement
ISTH DIC score	Only after a DIC-causing condition is identified. ≥5 = overt DIC ; <5 = nonovert. Not widely applied in obstetrics

General management

1. **Identify and remove the cause** (e.g., deliver) to stop defibrination.
2. **Vigorous restoration of circulation**: perfusion lets the liver and endothelium resynthesize procoagulants and lets the reticuloendothelial system **clear activated factors, fibrin and FDPs**.
3. **Rapid procoagulant replacement** with incisions or extensive lacerations and severe bleeding.
 - ⚠ **Heparin: abandoned.**
 - ⚠ **TXA and ε-aminocaproic acid: not first-line** in DIC: fibrinolysis is needed to dissolve the widespread fibrin thrombi.
 - **rFVIIa**: insufficient evidence for obstetrical coagulopathies.

4. Surgical Management of Hemorrhage

- Evidence quality is generally **poor** (AHRQ).
- 6660 PPH cases: invasive procedure **4%**, hysterectomy **1%**; conservative surgery or embolization **failed in 15%**.

- **Uterine balloon tamponade** for atony has **doubled** in a decade (Ch 42).
- **Topical hemostatic agents** for low-pressure oozing (mainly at cesarean hysterectomy).

Uterine compression sutures (B-Lynch "brace")

- **No. 2 chromic** suture compresses the anterior and posterior walls together; looks like **suspenders**. Several modifications exist.
- Mainly for **atony**; may reduce DIC bleeding while coagulation is corrected by transfusion.
- Success: B-Lynch **948 cases, 7 failures**; other series **20–25% failure**.
- Rare complications: **uterine ischemic necrosis with peritonitis**; later **uterine wall defects or synechiae**. Most later pregnancies are uneventful.

B-Lynch steps (Fig 44-4)

1. Enter the cavity through the lower segment **below** the hysterotomy (one side).
2. Exit **above** the incision; loop **over the fundus** to the back.
3. Re-enter the cavity through the **posterior wall** (at the level of the incision); cross to the other side inside.
4. Exit posteriorly; loop **over the fundus** to the front.
5. Re-enter **above** the incision.
6. Exit **below** the incision; **tie ends 1 and 6** below the incision; close the hysterotomy as usual.

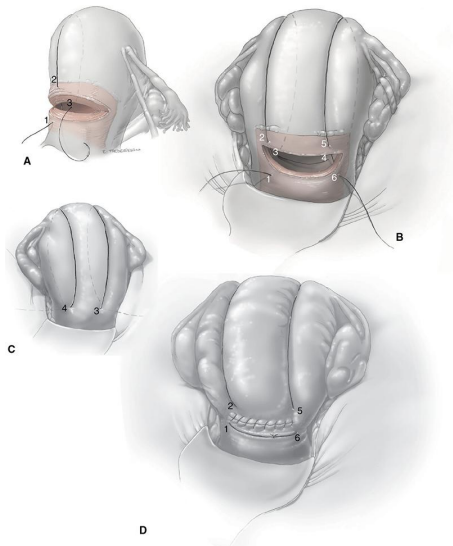


Figure 44-4. B-Lynch compression suture: numbers trace the path from front, over the fundus, through the posterior wall, and back over the fundus; tied below the hysterotomy.

Artery ligation

- Indications: **hysterotomy laceration, atony, abnormal placentation**.
- **Uterine artery ligation** (uni- or bilateral): mainly for **lacerations at the lateral hysterotomy**; less helpful for other causes. Suture passes through the lateral uterine wall anteriorly, curves posteriorly and re-enters anteriorly, encircling the uterine vessels.
- Torrential PAS bleeding: a **tourniquet around the lower segment** may slow bleeding during resuscitation and hysterectomy dissection.

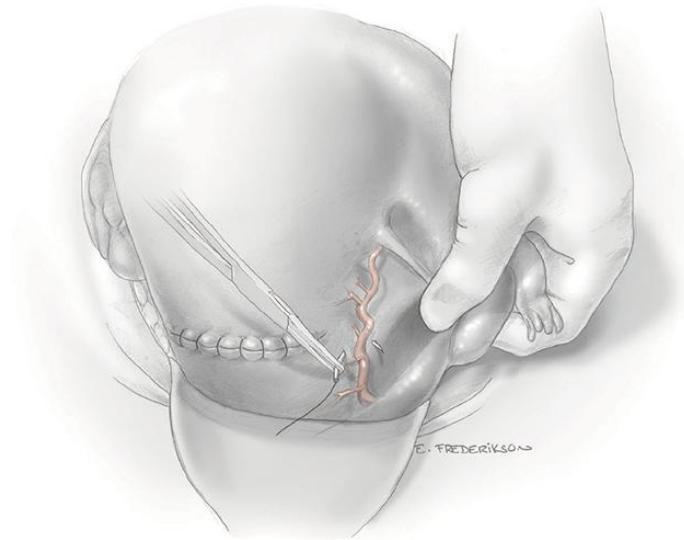


Figure 44-5. Uterine artery ligation: the suture passes through the lateral myometrium to encircle the ascending uterine artery.

Internal iliac (hypogastric) artery ligation

- Technically difficult; **successful in only ~half**. Helped with **placenta previa**; **not particularly helpful for atony**.
- **Technique:**
 1. Open the peritoneum over the common iliac and dissect to the **bifurcation** of external and internal iliacs.
 2. **Palpate pulses distal to the external iliac** (inguinal area) to confirm the correct vessel.
 3. Ligate **~5 cm distal to the common iliac bifurcation**, which avoids the **posterior division**.
 4. Incise the areolar sheath longitudinally; pass a **right-angle clamp beneath the artery from lateral to medial**.
 ⚠ Don't perforate the **internal iliac vein**.
 5. Tie with (usually **nonabsorbable**) suture.
- **Mechanism:** **~85% ↓ pulse pressure** distally → arterial system behaves like a **venous-pressure** system, so clot and pressure can stop the bleeding.
- **Fertility is preserved:** 17 women, 21 later pregnancies (13 normal, 3 miscarriages, 3 terminations, 2 ectopics).

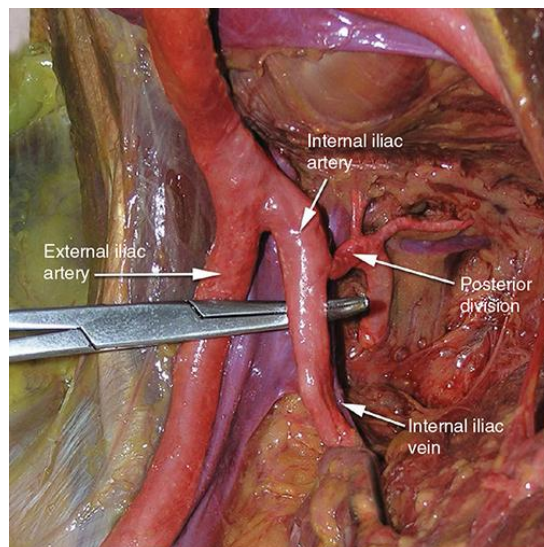


Figure 44-6. Right internal iliac ligation (cadaver): right-angle clamp under the anterior division, just distal to the posterior division; note the internal iliac vein beneath.

Angiographic embolization

- For intractable bleeding when surgical access is hard; use has **increased remarkably**.
- **Effective in ~90%** (>500 pregnancy cases). **Fertility is not impaired**.
- ⚠ **Hemodynamically unstable women with active bleeding should not leave the OR** for radiology.

- Rare but severe complications: iliac artery rupture, **uterine ischemic necrosis**, uterine infection, **buttock necrosis and paraplegia** (bilateral internal iliac embolization), myometrial defect in a later pregnancy.
- Preoperative balloon catheters for anticipated massive loss: see PAS (Ch 43).

Aortic compression

- **REBOA** (resuscitative endovascular balloon occlusion of the aorta): prophylactic use reported to reduce blood loss and hysterectomy (e.g., 11 PAS cases). ↓ distal perfusion, ↑ afterload, redistributes volume to heart and brain. ⚠ Prolonged occlusion → **limb ischemia and amputation**.
- **Manual aortic compression above the sacral promontory** for torrential pelvic bleeding; with the abdomen closed, press **above the umbilicus**.

Pelvic packing

- If **hysterectomy fails** to stop bleeding: pack the pelvis with **gauze rolls** and end the operation.
 - Temporizing before embolization, or left **48–72 h** and removed if the patient is stable.
 - Review of 104: **success 79%**; 24 failures, 13 of whom died → overall mortality **13%**.
- **Umbrella (parachute) pack**: sterile plastic bag filled with knotted gauze rolls placed transabdominally, **stalk out through the vagina**, traction from a **1-L fluid bag** hung over the foot of the bed; removed vaginally after **24 h**. Lifesaving when all else fails, especially in low-resource settings.

Quick-Recall: Blood Products and Hemostatic Agents

Product / agent	Key content	Effect / dose	Trigger / caution
Whole blood	RBC + plasma + fibrinogen (600–700 mg), no platelets	↑ Hct 3–4%/unit	Ideal ; fewer donor exposures; lacks V, VIII, XI when stored
PRBC	Hct 55–80%, minimal fibrinogen	↑ Hct 3–4%/unit	Transfuse if Hct <25% with ongoing bleeding
FFP	All factors incl. fibrinogen (600–700 mg)	10–15 mL/kg ; 30-min thaw	Fibrinogen <150 or ↑ PT/PTT; not a volume expander
Cryoprecipitate	200 mg fibrinogen , VIII, vWF, XIII, fibronectin	15–20 units (3–4 g) → +150 mg/dL	Most wasted product (~17%)
Fibrinogen concentrate	Virus-inactivated	1 g → +40 mg/dL	Congenital hypofibrinogenemia
Platelets	Apheresis bag = 6 units	+~20,000/μL per bag	Keep >50,000 when bleeding; D-neg → D-neg donors; sepsis risk
rFVIIa	Binds tissue factor	—	Ineffective if fibrinogen <50 or platelets <30,000 ; thrombosis; marginal benefit
TXA	Antifibrinolytic	1 g IV , within 3 h	WOMAN 1.2% vs 1.7% ; ACOG: not prophylaxis/first-line; not for DIC
MTP (Parkland)	Rounds of 5 RBC + 3 FFP ± 6-pack platelets ± cryo	Start after 4–5 units in ~2 h	Plasma:RBC ~1:1.4 best in trauma

Key Numbers to Memorize

- True blood loss **2–3×** the visual estimate; **3% Hct drop ≈ 500 mL**
- Urine output **≥30 mL/h (ideally ≥50)**
- Shock decompensates at **>25%** volume loss; **70%** of blood volume is in venules
- Crystalloid: **20%** stays intravascular at 1 h → give **2–3×** the estimated loss
- Antibody screen misses **0.03–0.07%**
- Cardiac output falls at **Hb ~7 / Hct ~20%**; Williams transfuses at **Hct <25%**

- Check coagulation after **~5 units**; platelets **>50,000**; fibrinogen **>150 mg/dL**; FFP **10–15 mL/kg**
- Whole blood shelf life **40 days**; 1 unit ↑ Hct **3–4%**
- Cryo **200 mg** fibrinogen/unit; fibrinogen concentrate **1 g → +40 mg/dL**
- rFVIIa useless if fibrinogen **<50** or platelets **<30,000**
- WOMAN TXA **1 g**: deaths **1.2% vs 1.7%**; give within **3 h**
- MTP after **4–5 units in 2 h**; plasma:RBC **1:1.4** best, **1:8** worst
- Hemolytic reaction **1 in 14,000**; TACO **~1%**; TRALI **≥1 in 12,000**; both within **6 h**
- Contaminated platelets **1 in 1000–2000**; HIV/HCV **1 in 1–2 million**; HBV **<1 in 100,000**
- DIC **0.03–0.35%** of pregnancies; sole cause of **0.2%** of deaths; HELLP DIC **~10%**
- Dead fetus: DIC in **~¼ after 1 month**; D&E DIC **1.6%**
- Fibrinogen late pregnancy **300–600**; clinical threshold **150**; severe **<50 mg/dL**; ISTH overt DIC **≥5**
- B-Lynch **no. 2 chromic**; failure **20–25%** in series
- Internal iliac ligation **5 cm** distal to the bifurcation; **85%** ↓ pulse pressure; works in **~½**
- Embolization **~90%** effective; pelvic packing **79%** success, **48–72 h**; umbrella pack out at **24 h**